MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW POWER SCHOTTKY TTL, HEX BUS DRIVERS WITH 3-STATE OUTPUTS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/322B, dated 20 October 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.1, line two: Delete "Three" and substitute "Two".

1.2.3, case outlines, add the following:

"2 C-2 (20 terminal, .350" x .350"), square chip carrier package.
X C-2A (20 terminal, .350" x .350"), square chip carrier package."

* 1.3, input voltage range, delete and substitute as follows:

"Input voltage range - - - - - - - - - - - - - - - - - - - -1.5 V to 7.0 V"

PAGE 2

1.3, thermal resistance, delete and substitute the following:

"Thermal resistance, junction-to-case (Θ_{JC}):

Cases E, F, X, and 2 - - - - - - - - - - - - - - See MIL-M-38510, appendix C"

1.3, Junction temperature, delete and substitute:

"Junction temperature (T_J) 3/ - - - - - - - - - - +175°C".

1.4, after supply voltage, add "(V_{CC})".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
3	3
4	4
7	7
8	8
9	REPRINTED WITHOUT CHANGE
10	10
12a	NEW PAGE

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.

3.2.4, last sentence: Delete "All manufacturers'" and substitute "All qualified manufacturers'".

Following footnote 2/, add new footnote as follows:

"3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883."

PAGE 5

Table II, delete class C devices column in its entirety.

Table II, entry for "Interim electrical tests": Delete "(pre burn-in)".

Table II, entry for "Group A electrical test, class B devices": Add "10, 11".

Table II, entry for "Additional electrical tests for group C inspections": Delete in its entirety.

3.6, second sentence: Delete in its entirety.

PAGE 6

4.2a (1), delete and substitute:.

"Test condition D, E, or F, using the circuit shown on figure 3, or equivalent."

4.2c, delete in its entirety and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

4.4.3b: Delete in its entirety.

4.4.3c (1), delete and substitute:

"Test condition D, E, or F, using the circuit shown on figure 3, or equivalent."

4.4.3c (3): Delete "appendix B of MIL-M-38510 and".

PAGE 12

Figure 4, under "IN PHASE OUTPUT": Add "01, 03".

Figure 4, under "OUT OF PHASE OUTPUT": Add "02, 04".

PAGE 13

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

* Table III, I_{IH2}, test 53 pin $\overline{E}1$, test 54 pin 1A, test 55 pin 2A, test 56 pin 3A, and test 57 pin 4A: Delete "5.5" and substitute "7.0" each place.

Table III, at end of table, add the following: "See footnotes at end of device type 01."

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:11Z
Check the source to verify that this is the current version before use.

PAGE 14

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

* Table III, I_{IH2}, test 58 pin 5A, test 59 pin 6A, and test 60 pin $\overline{E}2$: Delete "5.5" and substitute "7.0" each place.

Table III, I_{IL2}, test 68, $\overline{E}1$ column: Delete "0.4 V".

Table III, subgroup 10, test number 92, pin 3: Add "OUT".

PAGE 15

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

Table III, footnote 1/, I_{IL1}, circuit F: Delete "-160/-400" and substitute "0/-200".

* Table III, footnote 2/, I_{OS}, circuit C: Delete "-30/-130" and substitute "-40/-225".

Table III, add the following footnotes:

"5/ For cases 2 and X, pins not referenced are NC."

PAGE 16

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

* Table III, V_{IC}, test number 40, pin 4A, add "-18 mA".

* Table III, V_{IC}, test number 41, pin 4A, delete "-18 mA". Also, test number 41, pin 5A, add "-18 mA".

* Table III, V_{IC}, test number 42, pin 5A, delete "-18 mA". Also, test number 42, pin 6A, add "-18 mA".

* Table III, V_{IC}, test number 43, pin 6A, delete "-18 mA".

* Table III, I_{IH2}, test 53 pin $\overline{E}1$, test 54 pin 1A, test 55 pin 2A, test 56 pin 3A, and test 57 pin 4A: Delete "5.5" and substitute "7.0" each place.

Table III, at end of table, add the following: "See footnotes at end of device type 02."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.

PAGE 17

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

* Table III, I_{IH2}, test 58 pin 5A, test 59 pin 6A, and test 60 pin $\overline{E}2$: Delete "5.5" and substitute "7.0" each place.

Table III, I_{IL2}, test 68, $\overline{E}1$ column: Delete "0.4 V".

PAGE 18

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

* Table III, footnote 2/, I_{OS}, circuit C: Delete "-30/-130" and substitute "-40/-225".

Table III, add the following footnotes:

"5/ For cases 2 and X, pins not referenced are NC."

PAGE 19

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

* Table III, I_{IH2}, test 53 pin $\overline{E}1$, test 54 pin 1A, test 55 pin 2A, test 56 pin 3A, and test 57 pin 4A: Delete "5.5" and substitute "7.0" each place.

Table III, at end of table, add the following: "See footnotes at end of device type 03."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.

PAGE 20

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E1}$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E2}$	V_{CC}

"

* Table III, I_{IH2}, test 58 pin 5A, test 59 pin 6A, and test 60 pin $\overline{E2}$: Delete "5.5" and substitute "7.0" each place.

PAGE 21

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E1}$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E2}$	V_{CC}

"

Table III, footnote 1/, I_{IL1}, circuit F: Delete "-160/-400" and substitute "0/-200".

* Table III, footnote 2/, I_{OS}, circuit C: Delete "-30/-130" and substitute "-40/-225".

Table III, add the following footnotes:

"5/ For cases 2 and X, pins not referenced are NC."

PAGE 22

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E1}$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E2}$	V_{CC}

"

Table III, subgroup 1, test number 29, unit column: Delete "μA" and substitute "μ".

* Table III, I_{IH2}, test 53 pin $\overline{E1}$, test 54 pin 1A, test 55 pin 2A, test 56 pin 3A, and test 57 pin 4A: Delete "5.5" and substitute "7.0" each place.

Table III, at end of table, add the following: "See footnotes at end of device type 04."

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:11Z
Check the source to verify that this is the currentversion before use.

PAGE 23

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

* Table III, I_{IH2}, test 58 pin 5A, test 59 pin 6A, and test 60 pin $\overline{E}2$: Delete "5.5" and substitute "7.0" each place.

PAGE 24

Table III, case designators and pin designations: Delete and substitute the following:

"

Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Cases 2, X 5/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	$\overline{E}1$	1A	1Y	2A	2Y	3A	3Y	GND	4Y	4A	5Y	5A	6Y	6A	$\overline{E}2$	V_{CC}

"

* Table III, footnote 2/, I_{OS}, circuit C: Delete "-30/-130" and substitute "-40/-225".

Table III, add the following footnotes:

"5/ For cases 2 and X, pins not referenced are NC."

PAGE 25

6.1, delete in its entirety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.

The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1258)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V; V_{IN} = 0.7 V or 2.0 V; I_{OH} = -1 mA	All	2.4	---	V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V; V_{IN} = 2.0 V or 0.7 V; I_{OL} = 12 mA	All	---	0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; V_{IN} = -18 mA T_C = 25°C	All	---	1.5	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.7 V	All	---	20	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 7.0 V	All	---	0.1	mA
Inhibited-state output leakage current	I_{O1}	V_{CC} = 5.5 V; V_{IN} = 2.0 V; V_{OUT} = 2.4 V	All	---	20	µA
	I_{O2}	V_{CC} = 5.5 V; V_{IN} = 2.0 V; V_{OUT} = 0.4 V	All	---	-20	µA
Low-level input current at A input	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.4 V; $\overline{E}$ inputs at 0.4 V	All	0.0	-400	µA
Low-level input current at $\overline{E}$ input	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.4 V	All	-1.0	-400	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	All	-15	-225	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01 02 03 04	--- --- --- ---	24 21 24 21	mA mA mA mA
Propagation delay time (low-to-high level output)	t_{PLH}	V_{CC} = 5.0 V R_L = See figure 4 C_L = 50 pF	01 02 03 04	2 2 2 2	21 20 21 20	ns ns ns ns
Propagation delay time (high-to-low level output)	t_{PHL}	V_{CC} = 5.0 V R_L = See figure 4 C_L = 50 pF	01 02 03 04	2 2 2 2	29 23 29 23	ns ns ns ns
Propagation delay time (disabled to high-level output)	t_{PZH}	V_{CC} = 5.0 V R_L = See figure 4 C_L = 50 pF	01 02 03 04	2 2 2 2	45 45 45 45	ns ns ns ns

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

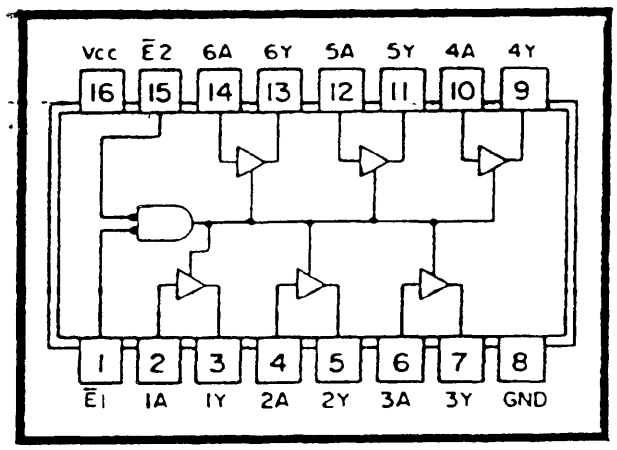
Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
Propagation delay time (high-level to disabled output)	t_{PHZ}	V_{CC} = 5.0 V, R_L = See figure 4 C_L = 50 pF	01	2	39	ns
			02	2	42	ns
			03	2	39	ns
			04	2	42	ns
Propagation delay time (disabled to low-level output)	t_{PZL}	V_{CC} = 5.0 V, R_L = See figure 4 C_L = 50 pF	01	2	52	ns
			02	2	58	ns
			03	2	52	ns
			04	2	58	ns
Propagation delay time (low-level to disabled output)	t_{PLZ}	V_{CC} = 5.0 V, R_L = See figure 4 C_L = 50 pF	01	2	45	ns
			02	2	45	ns
			03	2	45	ns
			04	2	45	ns

1/ Not more than one output should be shorted at a time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.

Device type 01

Cases E and F



Device type 02

Cases E and F

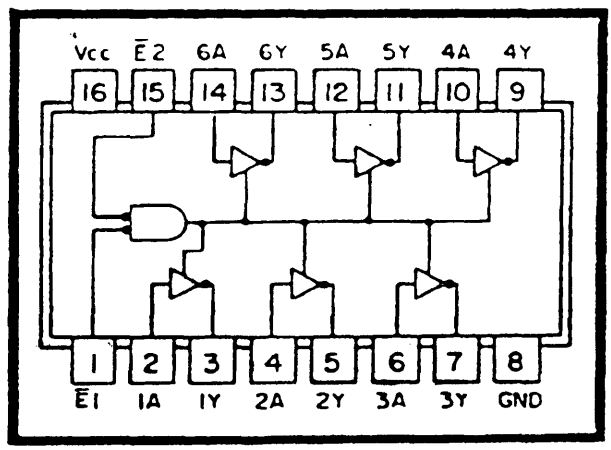


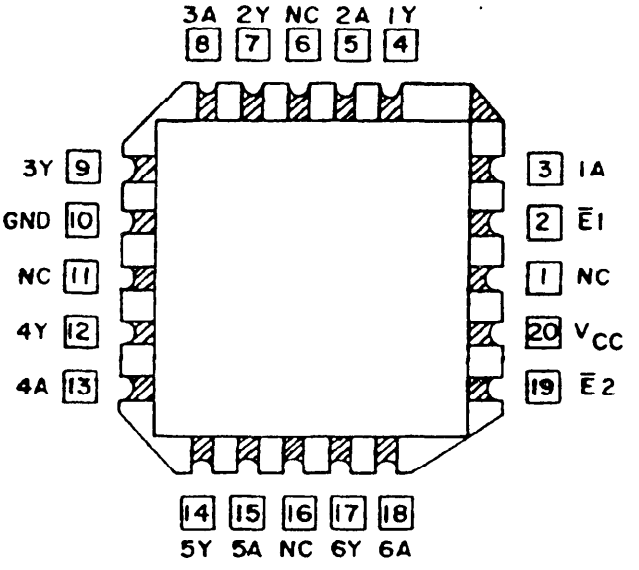
FIGURE 1. Terminal connections and logic diagram - top view

Supersedes page 7 of MIL-M-38510/322A
Amendment 3, of 13 September 1984.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.

Device types 01, 02, 03, and 04

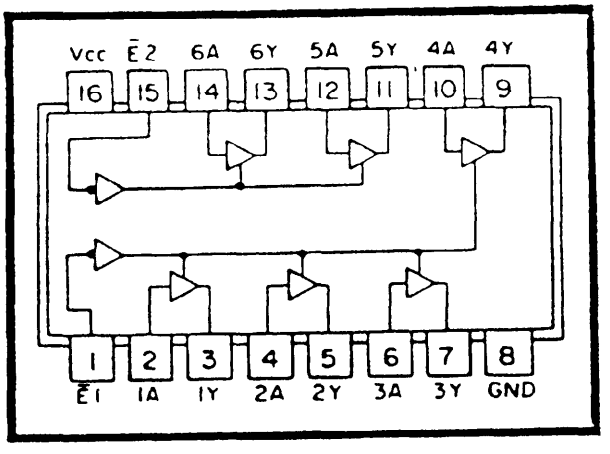
Cases X and 2



(TOP VIEW)

Device type 03

Cases E and F



Device type 04

Cases E and F

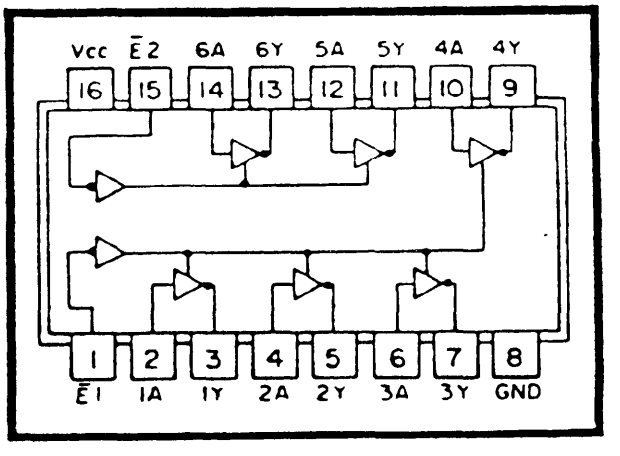


FIGURE 1. Terminal connections and logic diagram - top view - Continued.

Supersedes page 8 of MIL-M-38510/322A,
Amendment 3, of 13 September 1984.

Source: https://assist.dla.mil -- Downloaded: 202640-21T14:11Z
Check the source to verify that this is the current version before use.

Device type 01

Inputs			Outputs
$\overline{E}1$	$\overline{E}2$	A	Y
H	X	X	Z
X	H	X	Z
L	L	H	H
L	L	L	L

Device type 02

Inputs			Outputs
$\overline{E}1$	$\overline{E}2$	A	Y
H	X	X	Z
X	H	X	Z
L	L	H	L
L	L	L	H

Device type 03

Inputs		Outputs
$\overline{E}1$	A	Y
H	X	Z
L	H	H
L	L	L

Device type 04

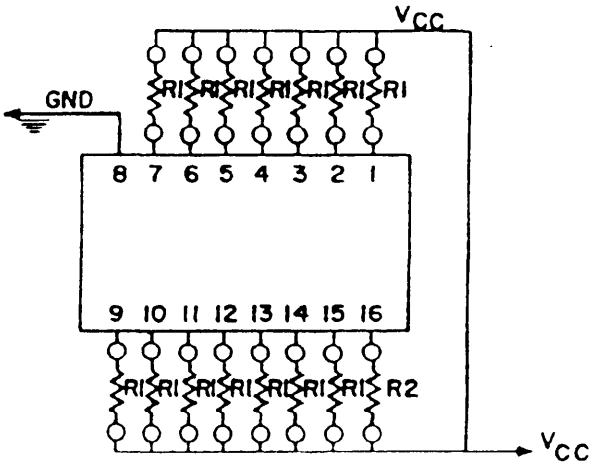
Inputs		Outputs
$\overline{E}1$	A	Y
H	X	Z
L	H	L
L	L	H

H = High level
L = Low level
X = Irrelevant
Z = High-impedance

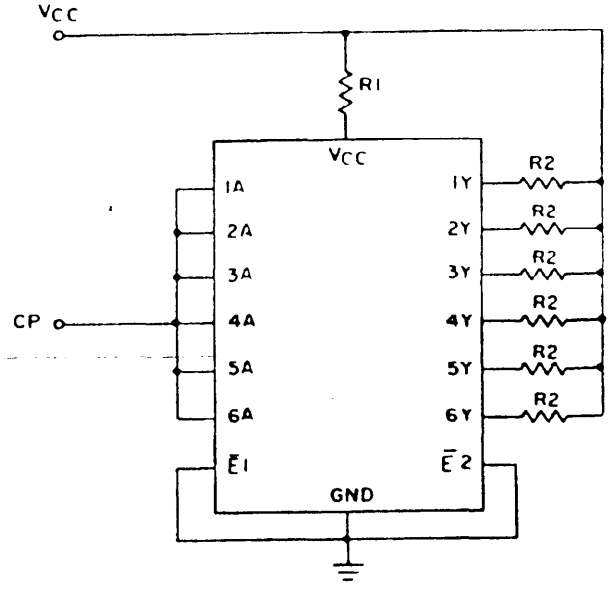
FIGURE 2. Truth table (each driver).

Reprinted without change.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.



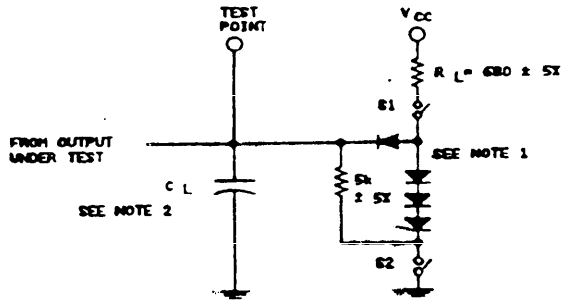
NOTES:
1. High temperature condition F(175°C) condition F burn-in circuit only.
2. R_1 = 2 kΩ.
3. V_{CC} = shall be chosen such that 5.0 V minimum is present at device terminal.



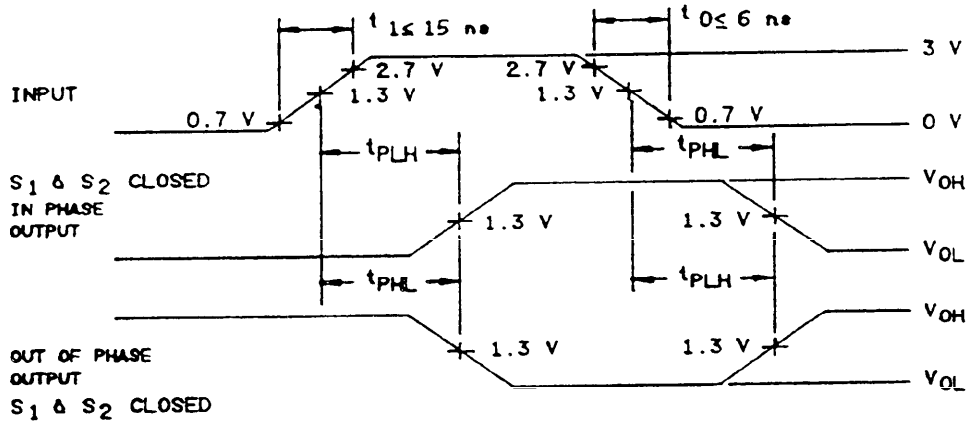
NOTES:
1. V_{CC} and R_1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R_2 = 470Ω ±5 percent.
3. CP = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 3. Burn-in and life test circuit.

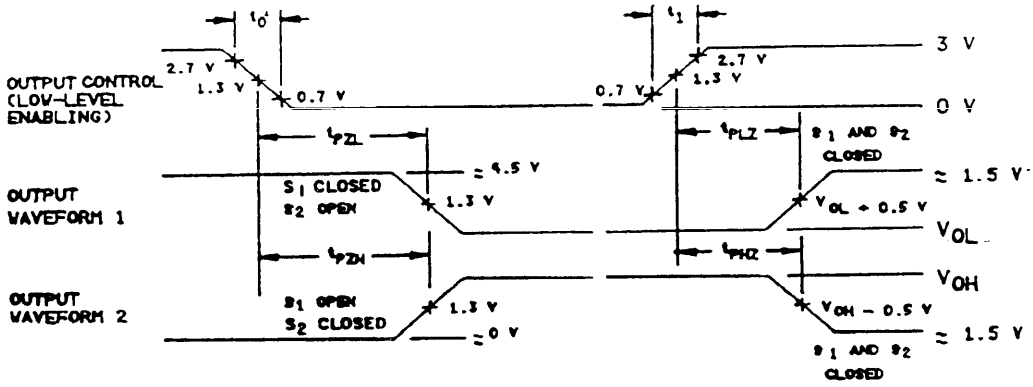
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:11Z
Check the source to verify that this is the current version before use.



ALTERNATE LOAD CIRCUIT



ALTERNATE VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES



ALTERNATE VOLTAGE WAVEFORMS
FOR TRI-STATE SWITCHING

NOTES:
1. Diodes are 1N3064 or equivalent.
2. C_L = 50 pF ±10 percent for t_{PLH}, t_{PHL}, t_{PZL}, and t_{PZH}; C_L = 15 pF minimum for t_{PHZ} and t_{PLZ}. C_L includes probe and jib capacitance.

FIGURE 4. Switching time test circuit and waveforms - Continued.

NEW PAGE

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:11Z
Check the source to verify that this is the current version before use.